AWS TASK LIST

S3

1. Configure ALB access logs storage in S3 bucket
2. Configure VPC Flow Logs storage in S3 bucket
3. Enable MFA Delete protection on S3 bucket
4. Enable Server Access Logging on S3 bucket
5. Generate pre-signed URLs for temporary access to S3 objects
6. Access S3 bucket securely from EC2 instance
7. Enable encryption on S3 bucket using SSE-S3/SSE-KMS
8. Create S3 bucket using AWS CLI
9. Mount S3 bucket on EC2 instance
10. Restore deleted objects using S3 Versioning
11. Configure private S3 bucket with CloudFront distribution
12. Configure lifecycle policy to automatically delete expired objects
13. Configure multipart upload for large S3 objects
14. Configure Intelligent-Tiering storage class for cost optimization
15. Configure S3 event notifications for object creation/deletion/restoration
16. Configure cross-account S3 bucket access
17. Configure S3 bucket access without NAT Gateway or Internet Gateway using VPC Endpoint

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EC2

1. Attach a single EBS volume to multiple EC2 instances
2. Create AMI from an EC2 instance hosting NGINX
3. Copy AMI from one AWS account to another AWS account
4. Recover and access key-loss EC2 instance
5. Change instance type of running EC2 instance
6. Extend root EBS volume on EC2 instance
7. Retrieve data from terminated EC2 instance
8. Retrieve metadata of Amazon Linux EC2 instance
9. Import custom SSH key pair into AWS
10. Configure EC2 monitoring for CPU, Memory, and Disk utilization
11. Configure EC2 email notifications for high CPU or Disk usage
12. Create auto-healing infrastructure using Auto Scaling

VPC & NETWORKING

1. Enable and analyze VPC Flow Logs
2. Configure VPC Endpoint and integrate with DynamoDB
3. Implement peering between 4 VPCs using Transit Gateway
4. Configure AWS VPN connection

5. Configure Site-to-Site VPN connection
6. Create and attach Elastic Network Interface (NIC/ENI)
7. Create highly available VPC architecture with public and private subnets
8. Implement secure VPC architecture using NAT Gateway
9. Configure VPC architecture using NAT Instance instead of NAT Gateway
10. Configure cross-account VPC peering architecture
11. Explore difference between Gateway Endpoint and Interface Endpoint

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LOAD BALANCER & AUTOSCALING

1. Configure Application Load Balancer with backend port changes
2. Configure Auto Scaling with cooldown time of 5 minutes
3. Configure Auto Scaling with warm-up time of 2 minutes
4. Configure Auto Scaling with Load Balancer and Dynamic Scaling Policy
5. Configure weighted routing policy (40-40-20) on Load Balancer
6. LOAD BALANCER WITH AUTOSCALING WITH THE RULE 70-30 WEIGHTED POLICY
7. Implement NGINX as a Load Balancer
8. Configure path-based routing on Load Balancer (Laptop/Mobile/Home)
9. Configure ALB with Route53, ACM, and WAF
10. Configure ALB access logs and store them in S3 bucket
11. Configure Network Load Balancer with Proxy Protocol
12. Configure Session Affinity and Connection Idle Timeout
13. Configure Target Group Stickiness
14. Implement Cross-Account Network Load Balancer architecture
15. Implement Blue-Green Deployment architecture on AWS

CLOUDWATCH & MONITORING

1. Monitor Access Logs and Error Logs using CloudWatch Agent
2. Configure custom CloudWatch metrics for memory utilization
3. Configure centralized logging using CloudWatch
4. Configure CloudWatch alarms for CPU, memory, and disk utilization
5. Configure SNS email notifications for infrastructure alerts
6. Analyze logs using CloudWatch Logs Insights
7. Configure monitoring dashboard for EC2 and infrastructure metrics
8. Configure email notifications for IAM policy and role changes
9. Configure email notifications for S3 bucket and bucket policy changes

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LAMBDA

1. Create EBS snapshot using Lambda
2. Delete EBS snapshots using Lambda
3. Integrate S3, Lambda, and DynamoDB

4. Configure S3 trigger using Lambda
5. Schedule automatic start and stop of EC2 instances using Lambda
6. Automate snapshot creation and cleanup using Lambda
7. Filter EC2 instances running more than one hour and send email notification

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IAM & SECURITY

1. Configure AWS Certificate Manager (ACM)
2. Create IAM roles with time-limited permissions for EC2 instances
3. Configure cross-account access using IAM Roles
4. Create and configure AWS WAF
5. Configure NGINX reverse proxy with SSL certificate
6. Configure IAM users with CLI-only access
7. Retrieve temporary credentials using EC2 Instance Metadata Service
8. Monitor IAM changes using CloudTrail and SNS

LINUX / WEB SERVER

1. Configure LAMP Stack
2. Configure NGINX as Reverse Proxy
3. Host static website using NGINX with mounted directory (/mnt)
4. Host NGINX and HTTPD services on a single server
5. Configure Reverse Proxy architecture using NGINX
6. Configure SSL-enabled NGINX server
7. Perform web server setup on Ubuntu and Amazon Linux

DATABASE

1. Export DynamoDB table to S3 bucket
2. Launch RDS instance and create two database users
3. Configure RDS Multi-AZ deployment
4. Configure RDS automatic failover
5. Configure RDS Read Replicas
6. Monitor database replication lag and performance metrics
7. Import Excel data into DynamoDB and export DynamoDB data into Excel

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TASKS

1. Connect EC2 instance using VPC Endpoint without internet access
2. Create private S3 bucket accessible only from EC2 instance
3. Configure ALB with path-based routing

4. Create auto-healing infrastructure using Auto Scaling
5. Deploy NGINX with Reverse Proxy architecture
6. Implement secure VPC architecture using NAT Gateway
7. Configure centralized logging using CloudWatch
8. Create highly available web application architecture
9. Deploy three-tier architecture in private subnets
10. Deploy WAR application in Elastic Beanstalk
11. Configure CloudFront with SSL, Geo-Restriction, and real-time logs
12. Implement disaster recovery using cross-region snapshot copy
13. Configure production-grade monitoring and alerting system

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MIGRATION

1. CREATE A DATABASE IN UBUNTU ON VIRTUALBOX (LOCAL MACHINE) AND MIGRATE THESE DATABASE IN AWS RDS
2. Migrate on premises virtual machine (UBUNTU) on cloud EC2
